Practice workbook · Explain eye defects and how they are corrected or treated

Short sight, long sight, astigmatism, cataracts and corneal disease: where the image falls, cause, effect on vision, lenses, surgery and transplant risks.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EYE-DEFECTS-AND-TREATMENT

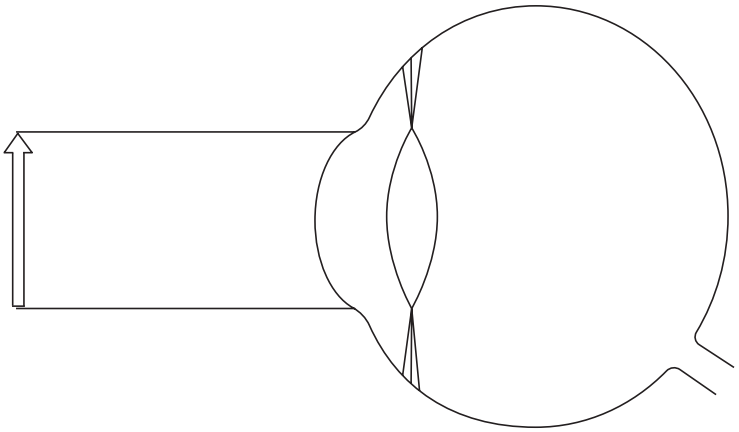
1. Question 8 · part b.iii · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.iii only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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1. Question 8 · part b.iii · 2 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.iii only.

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(iii) Describe a method of treating long sight.

(2)

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(c) Corneal transplants can be used to treat eye defects.

Suggest two risks associated with corneal transplants.

(2)

1 .....

2 .....

(Total for Question 8 = 13 marks)

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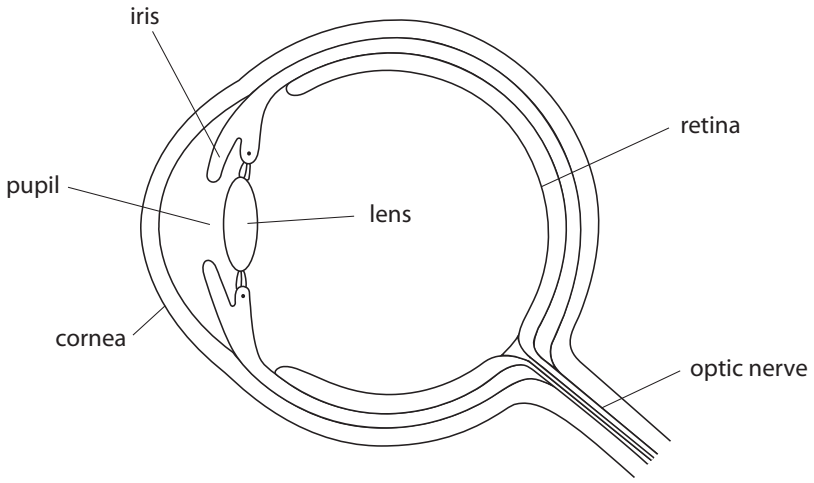
Working and answer

2. Question 3 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

3 The human eye contains structures that help to focus light on the retina.



(a) Which structures help to focus light on the retina? **(1)**

- ☒ **A** cornea and lens
- ☒ **B** iris and cornea
- ☒ **C** iris and pupil
- ☒ **D** pupil and lens

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2. Question 3 · part c · 2 marks

June 2019 Paper 1

Further practice · Attempt part c only.

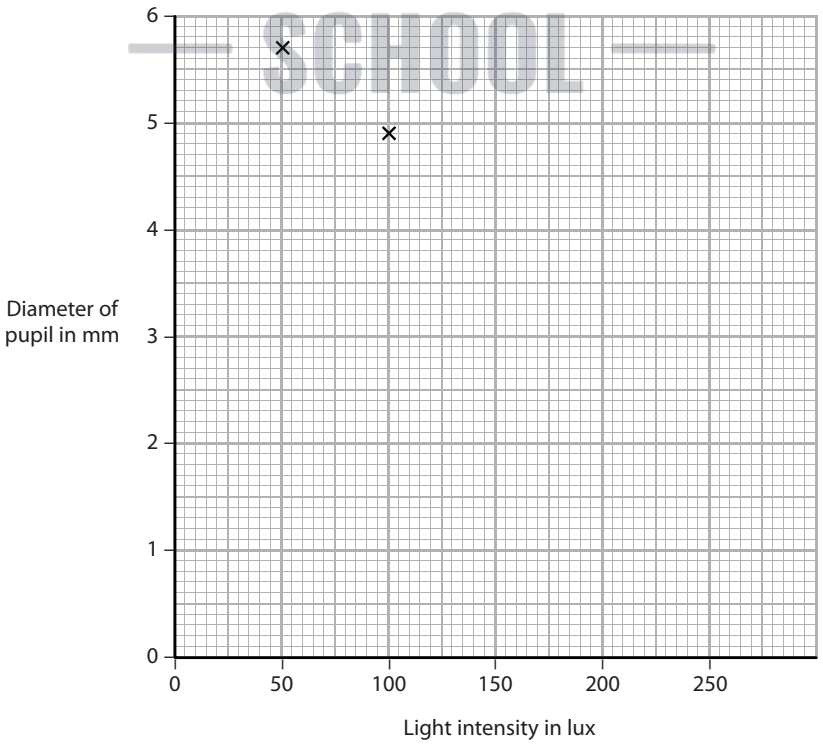
(b) The intensity of light entering the eye can be measured in a unit called lux.

The table shows the results of an investigation on how light intensity affects the diameter of the pupil in the eye.

Light intensity in lux	Diameter of pupil in mm
50	5.7
100	4.9
150	2.7
200	3.2
250	2.3

(i) Complete the graph by plotting the results for 150, 200 and 250 lux. (1)

(ii) Draw a straight line of best fit. (1)



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2. Question 3 · part c · 2 marks

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Further practice · Attempt part c only.

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(iii) Explain which result is anomalous.

(2)

(iv) State the effect of light intensity on the diameter of the pupil.

(1)

(c) Short-sightedness is a common eye condition.

Explain how short-sightedness affects vision.

(2)

(Total for Question 3 = 8 marks)

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2. Question 3 · part c · 2 marks · Working and answer

Working and answer

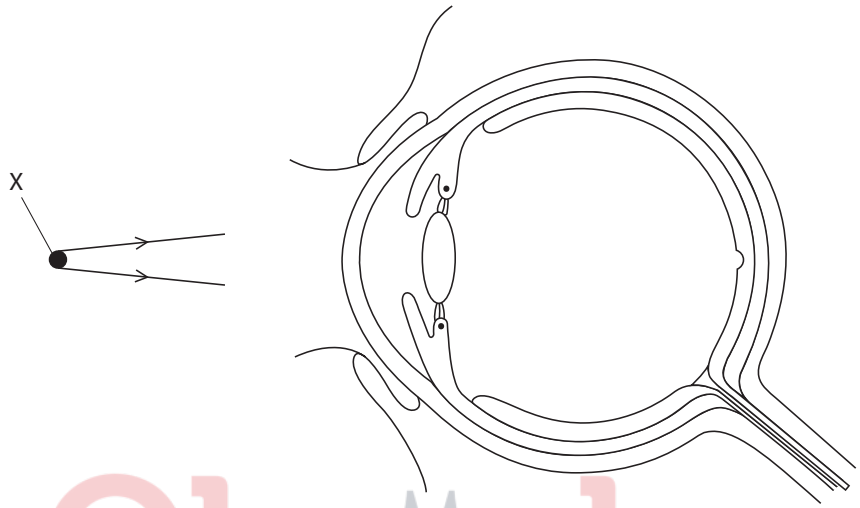
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3. Question 1 · part c · 3 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

1 The diagram shows a section through an eye.



(a) The table lists descriptions of parts of the eye.

Use the letters, A, B, C, D and E, to label the parts of the eye on the diagram.

(5)

Description of part	Letter
the layer that contains light sensitive cells	A
the muscles that change lens shape	B
the structure that controls the amount of light entering the eye	C
the structure that carries nerve impulses away from the eye	D
the structure that protects the eye from dust and grit	E

(b) A person looks at object X.

Complete the diagram by continuing the rays of light to show how the image of object X is formed on the retina.

(3)

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3. Question 1 · part c · 3 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

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(c) Name three defects of the eye that can be corrected by using glasses.

(3)

1 .....

2 .....

3 .....

(Total for Question 1 = 11 marks)

Working and answer

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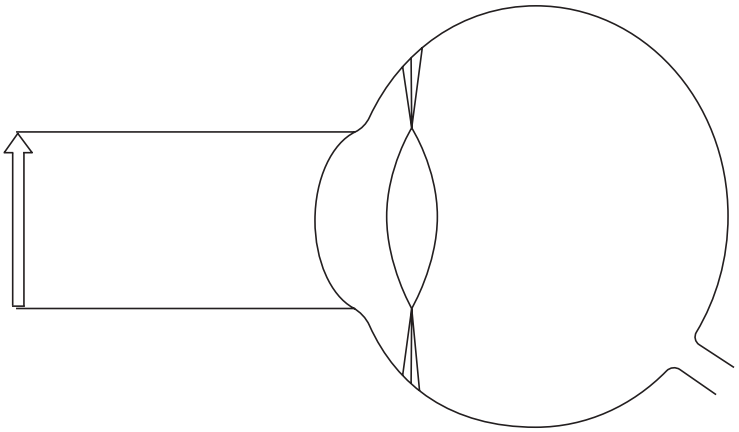
4. Question 8 · part b.i · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.i only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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4. Question 8 · part b.i · 1 mark · Working and answer

Working and answer

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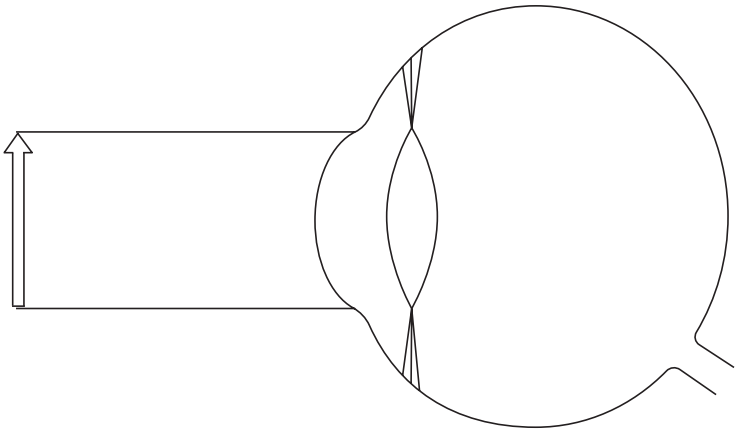
5. Question 8 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

(b) (i) State where the image of the arrow would be formed for someone with the condition of long sight.

(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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5. Question 8 · part b.ii · 1 mark

November 2021 Paper 1

Further practice · Attempt part b.ii only.

(iii) Describe a method of treating long sight.

(2)

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(c) Corneal transplants can be used to treat eye defects.

Suggest two risks associated with corneal transplants.

(2)

1 .....

2 .....

(Total for Question 8 = 13 marks)

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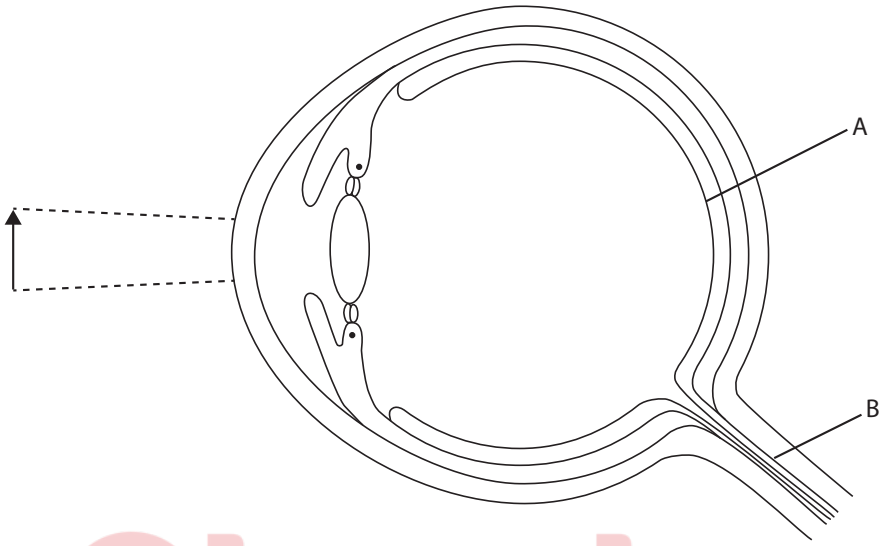
Working and answer

6. Question 6 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

6 The diagram shows a section through the eye.



(a) (i) Describe the functions of structures A and B. **(4)**

A .....

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B .....

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(ii) Complete the diagram to show how an image of the arrow is formed at the back of the eye.

(4)

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6. Question 6 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

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(b) Describe a simple experiment to show that stereoscopic vision would help a person to catch a ball thrown towards them.

(4)

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(c) Describe what is meant by the term **astigmatism**.

(2)

(Total for Question 6 = 14 marks)

6. Question 6 · part c · 2 marks · Working and answer

Working and answer

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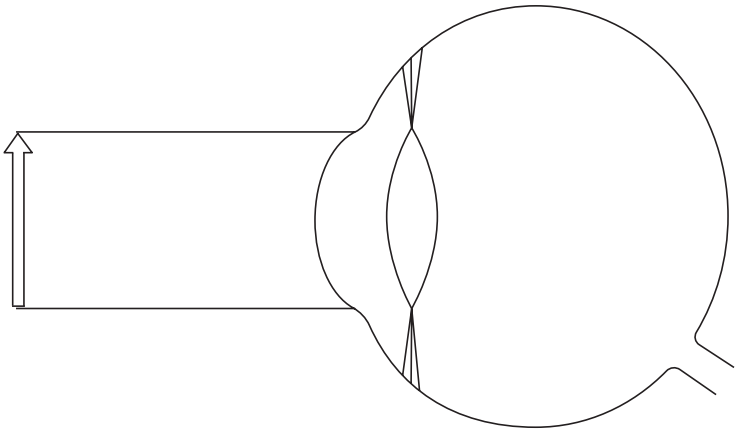
7. Question 8 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

8 The diagram shows a section through the eye.

The arrow is a near object.



(a) (i) Complete the diagram by continuing the rays from the arrow to show how the image of the arrow appears on the retina.

(5)

(ii) Describe the function of the optic nerve in allowing a person to see the image of the arrow.

(2)

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(1)

(ii) Give a feature of the eye that can lead to long sight.

(1)

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7. Question 8 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

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(iii) Describe a method of treating long sight.

(2)

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(c) Corneal transplants can be used to treat eye defects.

Suggest two risks associated with corneal transplants.

(2)

1 .....

2 .....

(Total for Question 8 = 13 marks)

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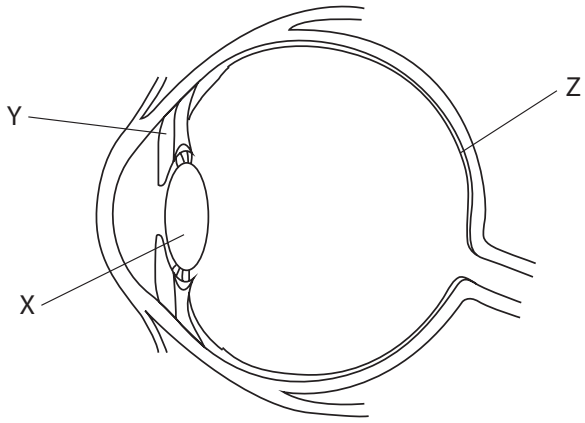
Working and answer

8. Question 1 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
structure X	refracts light rays
structure Y	protects the eye from pathogens
structure Z	detects light rays
	controls the shape of the lens
	transmits electrical impulses to the brain
	controls the size of the pupil

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8. Question 1 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

.....

.....

.....

(ii) State one treatment for cataracts.

(1)

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

8. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

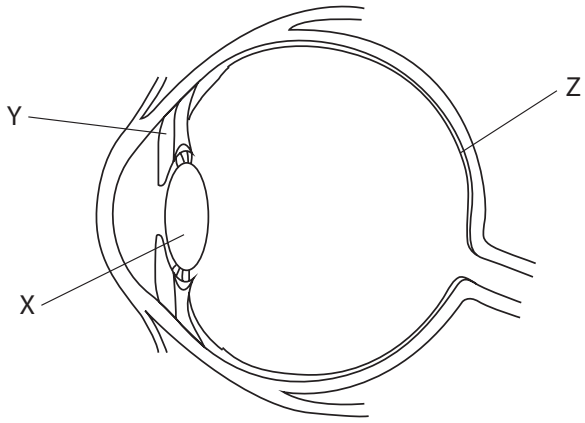
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9. Question 1 · part b.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.ii only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
structure X	refracts light rays
structure Y	protects the eye from pathogens
structure Z	detects light rays
	controls the shape of the lens
	transmits electrical impulses to the brain
	controls the size of the pupil

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9. Question 1 · part b.ii · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.ii only.

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Healthy eye Eye with cataract

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(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

9. Question 1 · part b.ii · 1 mark · Working and answer

Working and answer

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10. Question 3 · part c · 1 mark

June 2024 Paper 2

Further practice · Attempt part c only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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10. Question 3 · part c · 1 mark

June 2024 Paper 2

Further practice · Attempt part c only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

10. Question 3 · part c · 1 mark · Working and answer

Working and answer

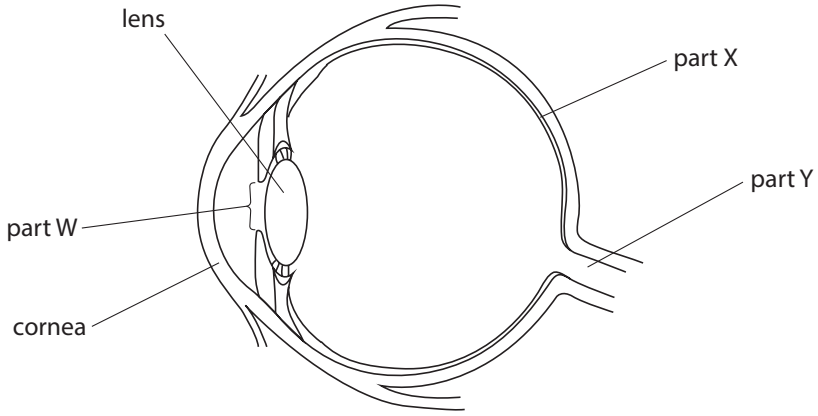
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11. Question 3 · part a.iv · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iv only.

3 (a) The diagram shows the human eye.



(i) Name part W and part X.

(2)

W .....

X .....

(ii) Describe the function of part Y.

(2)

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(iii) Explain the change in the shape of the lens when the eye views a distant object.

(2)

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11. Question 3 · part a.iv · 2 marks

November 2024 Paper 1

Further practice · Attempt part a.iv only.

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(iv) Describe how corneal transplants help to restore normal vision for people with corneal disease.

(2)

(b) The iris reflex controls the amount of light entering the eye.

Describe a method that could be used to investigate how the eye responds to different light intensities.

(4)

(Total for Question 3 = 12 marks)

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11. Question 3 · part a.iv · 2 marks · Working and answer

Working and answer

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